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Assessment number & title	Group Business Analytics Project Report and Demonstration
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RavenStack Customer Retention and Revenue Analytics

1.0 Executive Summary

This project applies business analytics to evaluate customer retention and revenue performance at RavenStack, a simulated Software-as-a-Service (SaaS) company. The company completed a pilot program involving 500 customer accounts and collected data on customers, subscriptions, feature usage, support activities and churn.

The main business problem is customer churn and its impact on recurring revenue. RavenStack recorded 110 churned accounts from 500 total accounts, representing a 22% churn rate. Microsoft Power BI was used to develop descriptive, diagnostic and predictive analytics dashboards to understand current performance, investigate churn factors, identify high-risk customers and forecast Monthly Recurring Revenue (MRR).

The analysis found that DevTools had the highest industry churn rate at 30.97%, while customers associated with escalated support cases recorded a 25.32% churn rate compared with 21.72% for non-escalated cases. Power BI Key Influencers also identified DevTools and Event referral as important churn-associated characteristics. The predictive analysis identified 78 active accounts as high risk and produced a three-month MRR forecast.

Based on these findings, we recommend that RavenStack prioritise high-risk active customers, investigate the DevTools customer experience, strengthen follow-up for escalated support cases and review Event-acquired customers. These actions can support more targeted customer retention and better decision-making before the public launch.

2.0 Organization and Business Context

RavenStack is an artificial intelligence (AI) powered SaaS product in stealth mode, it is designed to provide digital team members AI-powered productivity and collaboration capabilities. As such, RavenStack uses a B2B subscription business model, customers subscribe to one of three plans: basic, pro, or enterprise. Customers may pay either by the month or annually depending on their plan.

Prior to publicly launching RavenStack as a product, the company operated a pilot program that involved graduating coding boot camp students and initial stage technology companies. Data was collected during this time regarding the number of customer accounts created, total number of subscriptions purchased, feature utilization rates, number of support requests submitted, and number of customers who churning from the system (customer turnover).

Customer retention is critical to a subscription-based business like RavenStack's since its continued success will depend upon the recurring revenue generated by its subscribers. Therefore, management at RavenStack will need to assess how well its customers are performing, identify trends related to customer churn, and be aware of potential areas of risk related to customer retention. By using business analytics, RavenStack's management can convert the pilot data into meaningful information to help guide those decisions prior to going public with the product (IBM, n.d.)

3.0 Business Problem and Objectives

Although RavenStack was able to add 500 new customers in its initial pilot, the company is experiencing a high amount of customer churn. The descriptive analysis found that there were 110 churned customers and 390 customers still active. As such, the total churn percentage for RavenStack is approximately 22 percent. Additionally, the customer churn was directly tied to significant amounts of monthly recurring revenue (MRR) lost due to those customers who have churned. Because losing these types of customers could ultimately harm the ability of RavenStack to generate recurring revenue and decrease the return on investment related to the cost of adding new customers, this represents a major business issue for RavenStack.

Therefore, while the primary management issue does not exist solely in that some customers are churning, it does exist in the fact that RavenStack's management team will need to determine where the greatest risks of future churn lie. Furthermore, prior to the public launch of the product and service offering by RavenStack; they will need to identify what areas of their organization require the most focus regarding the development of strategies intended to minimize the number of customers lost.

The project has four objectives:

- 1: Evaluate RavenStack's overall customer and revenue performance by measuring total accounts, active accounts, churned accounts, churn rate and MRR.

- 2: Investigate customer characteristics and behaviours associated with churn, including industry, plan tier, feature usage and acquisition source.
- 3: Assess whether customer-support factors, particularly resolution time and escalation, are associated with retention outcomes.
- 4: Identify active customers with higher-risk characteristics and provide evidence-based management recommendations for improving retention before public launch.

4.0 Dataset and Analytical Approach

4.1 Dataset Description

The project used a simulated RavenStack SaaS dataset obtained from Kaggle. The dataset contains information relating to customer accounts, subscriptions, product usage, support activities and churn events. These different areas were important because the project needed to investigate churn from both customer and operational perspectives.

Five datasets were imported into Power BI.

Table 1: RavenStack Dataset Structure

Dataset	Main information	Purpose in the analysis
ravenstack_accounts	Account details, industry, plan tier, referral source and churn status	Customer segmentation and churn analysis
ravenstack_subscriptions	Subscription information, billing frequency and MRR	Revenue and subscription analysis
ravenstack_feature_usage	Feature usage and error counts	Product engagement and system-error analysis
ravenstack_support_tickets	Support priority, resolution time and escalation	Customer-support analysis
ravenstack_churn_events	Churn events and reason codes	Analysis of cancellation reasons

Using several related tables allowed the analysis to examine the business problem from different perspectives instead of relying only on the churn status stored in the accounts table.

4.2 Data Preparation and Modelling

The datasets were imported into Microsoft Power BI and prepared before visual analysis. Appropriate data types were checked for identifiers, numerical values and date fields. The tables were then connected through relevant identifiers such as account and subscription IDs so that information from different business activities could be analysed together.

The resulting model connected account information with subscription, feature usage, support and churn-event information.

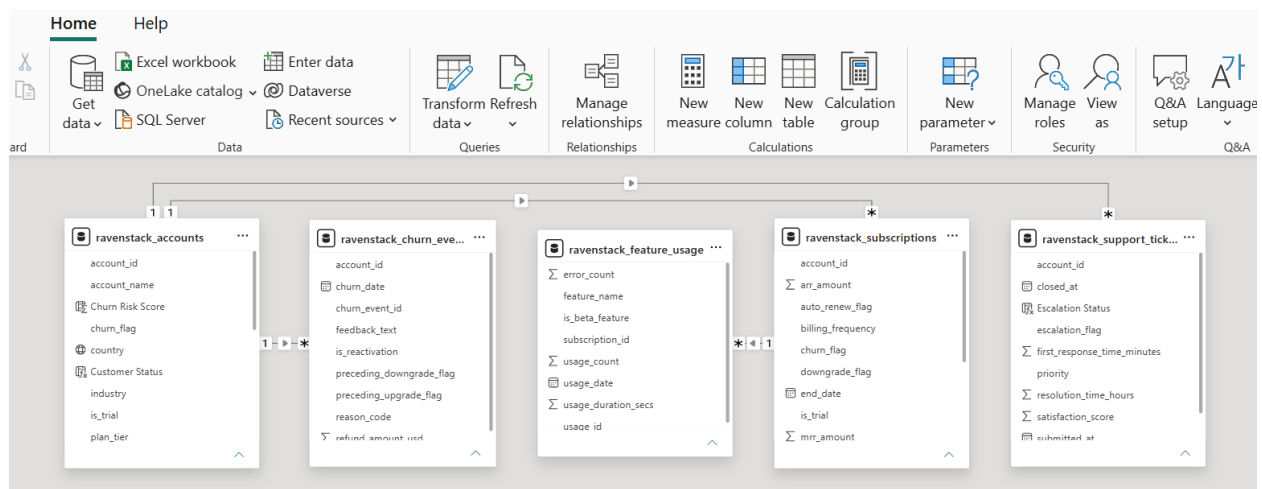


Figure 1. RavenStack Power BI Data Model

Measures and calculated fields were created to support the analysis. These included Total Accounts, Active Accounts, Churned Accounts, Churn Rate, Lost MRR and other measures required for diagnostic comparisons. Additional fields were created for the predictive analysis, including Churn Risk Score and Risk Level.

A Year Month date field was also created from subscription start dates. This allowed historical MRR to be aggregated into monthly periods rather than individual dates, making the data suitable for the three-month forecasting visual.

4.3 Analytical Approach

The analysis followed a three-page dashboard structure. The first page applied descriptive analytics to establish the current customer and revenue situation. The second page applied

diagnostic analytics to investigate possible churn factors. The third page applied predictive analytics to identify future customer risk and forecast MRR.

This structure was selected because it creates a logical decision-making process:

What happened? → Why did it happen? → What may happen next? → What should management do?

The first three questions are answered through the Power BI dashboards, while the final question is addressed through the decision-making implications and recommendations in this report.

5.0 Relevant Business Analytics Concepts

Business analytics utilizes data, analytical techniques, and visualizations to understand a company's performance as well as support decision making. We utilized four types of analytics in this project: descriptive; diagnostic; predictive; prescriptive. The four types together answer four management questions: "what happened?" (descriptive), "why did it happen?" (diagnostic), "what may happen next?" (predictive) and "what should we do?" (prescriptive).

5.1 Descriptive Analytics

Descriptive analytics summarises historical data to explain what has happened in a business using measures, percentages and visualisations (IBM, n.d.). In RavenStack, we used descriptive analytics to examine total accounts, active and churned accounts, churn rate, lost MRR, revenue distribution, customer acquisition channels and geographical usage.

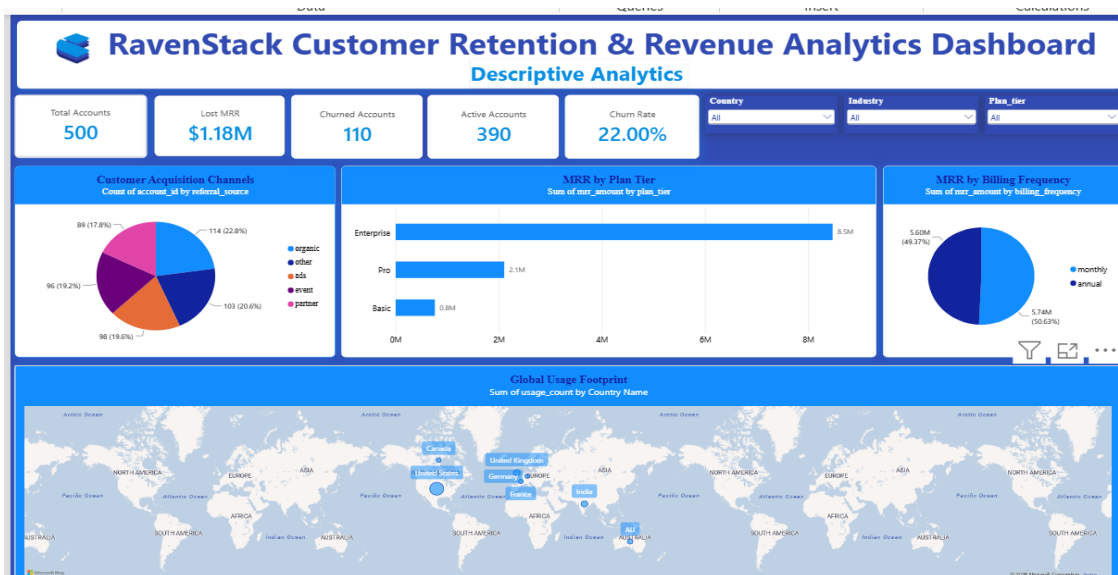


Figure 2. RavenStack Descriptive Analytics Dashboard

As shown in Figure 2, the dashboard provides management with an overview of customer and revenue performance and establishes the starting point for further investigation.

5.2 Diagnostic Analytics

Diagnostic analytics investigates why an outcome may have occurred by comparing variables and identifying relationships or patterns (Aquino & Jonker, 2025). For RavenStack, we examined churn across industry, plan tier, feature engagement, support resolution time, system errors, churn reasons and support escalation.

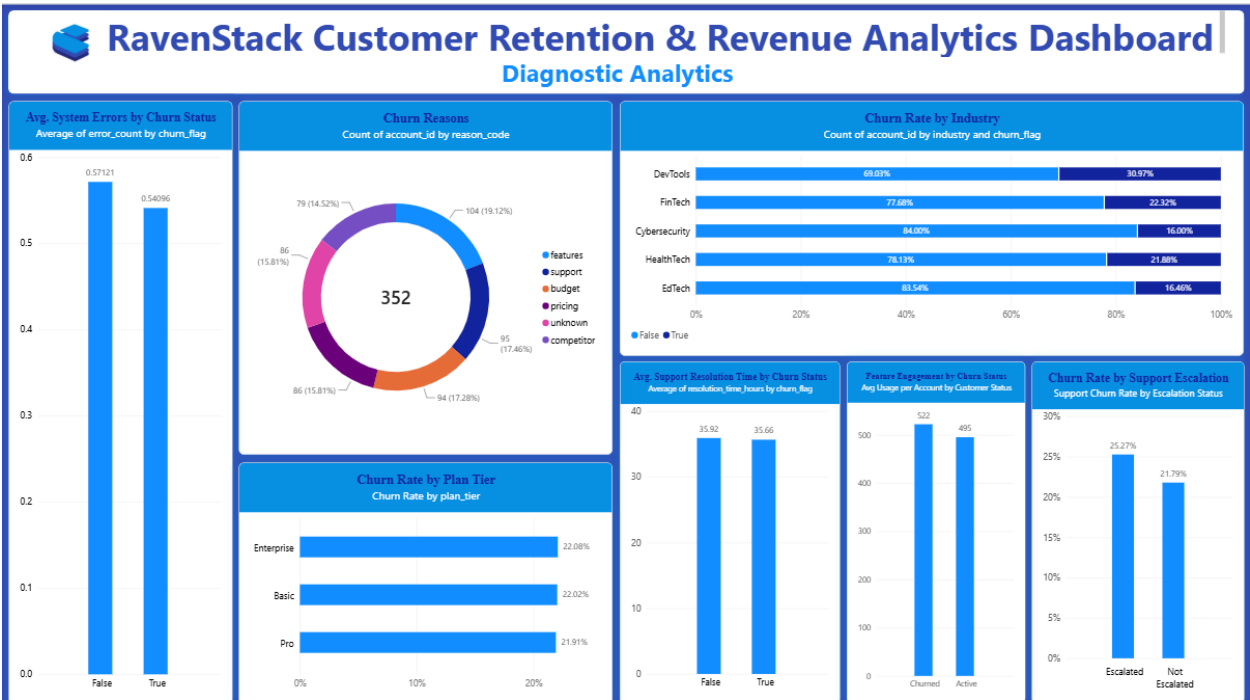


Figure 3. RavenStack Diagnostic Analytics Dashboard

The dashboard helped us determine which suspected churn factors showed meaningful differences. These findings are treated as associations rather than proof of causation.

5.3 Predictive Analytics

Predictive analytics uses historical data and identified patterns to estimate possible future outcomes (Finn & Smalley, n.d.). In RavenStack, we applied predictive analytics to customer churn risk and future MRR.

Power BI Key Influencers was used to identify characteristics associated with churn. These findings supported a rule-based risk classification of active customers into Low, Medium and High Risk. Historical monthly MRR was also used to produce a three-month forecast with a 95% confidence interval.



Figure 4. RavenStack Predictive Analytics Dashboard

The predictive dashboard helps management identify customers requiring proactive attention and provides a forward-looking view of recurring revenue.

5.4 Prescriptive Analytics

Prescriptive analytics focuses on what an organisation should do based on analytical evidence (Stryker, 2024). In this project, prescriptive analytics is represented through the decision-making implications and recommendations in Sections 7 and 8. It converts the descriptive, diagnostic and predictive findings into practical actions for RavenStack management.

6.0 Findings and Visualisations

6.1 Descriptive Findings: Current Customer and Revenue Performance

The descriptive analysis identified a clear retention problem during RavenStack's pilot program.

The company acquired 500 customer accounts, but only 390 remained active, while 110 had churned. This represents an overall churn rate of 22%.

Table 2: RavenStack Customer Status Summary

Performance Measure	Result
Total Accounts	500
Active Accounts	390
Churned Accounts	110
Churn Rate	22.00%

From a manager's viewpoint, we will view the success of RavenStack's pilot not just by achieving the goal of getting 500 accounts. We would also consider retention as a key measure for evaluating our success. If there is continuing high levels of customer loss, then it would diminish the long-term value generated from each additional customer gained.

The dashboard tracks both the monthly recurring revenue (MRR) lost when customers leave, and the MRR attributable to those who have left. Thus, customer churn is directly tied to financial performance. The distribution of recurring revenue (i.e., which plan tier or frequency of billing generates the most revenue), and how these factors vary over time are analyzed as well.

Customer Acquisition Channels visual illustrates how many customers came from different areas such as organically, via advertising, events, partners, etc. As a descriptor alone, this provides a general idea of where customers originally came from. Acquiring new customers from certain channels can become much more meaningful later in the process. Specifically, it was determined that event referrals were an influence on churn based upon the predictive analysis.

In total, the descriptive analysis provides an answer to the first managerial question: "What is going on?" RavenStack has been successful at gaining new customers and increasing its recurring revenue; however, since the overall customer churn rate is 22%, there is need for managerial action to improve customer retention.

6.2 Diagnostic Findings: Factors Associated with Churn

The diagnostic analysis investigated whether churn was concentrated within particular customer

groups or associated with product and support experiences.

6.2.1 Churn by Industry

Industry produced one of the clearest differences in the diagnostic analysis.

Table 3: Churn Rate by Industry

Industry	Churn Rate
DevTools	30.97%
FinTech	22.22%
HealthTech	21.88%
EdTech	16.46%
Cybersecurity	16.00%

DevTools recorded the highest churn rate at 30.97%, compared with only 16.00% for Cybersecurity. This difference suggests that RavenStack's retention problem is not equally distributed across all industries.

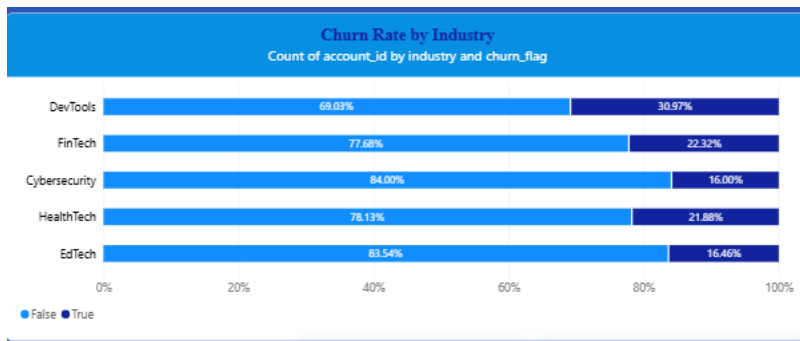


Figure 5. Churn Rate by Industry

As RavenStack's management team, we would make DevTools a priority segment for further investigation. Rather than immediately applying a general retention campaign to all customers, we would first investigate whether DevTools customers have different product requirements, onboarding expectations or feature needs.

6.2.2 Churn by Plan Tier

Plan tier created a different result. Enterprise experienced about 22.08% churn, Basic experienced about 22.00% churn and Pro experienced about 21.91% churn.

Therefore, the results indicate that there is no significant difference in the level of customer retention problems with each plan.

This result will be beneficial to our decision making process at this time since one potential reason that may have caused customer churn was the price or type of subscription the customers were using. Since the business problem indicated that pricing or subscription format could be causing some churn, as members of RavenStack's management team, we wouldn't make any drastic changes to one plan based on its churn rate. This is due to the fact that the churn rate for each of the three tiers are virtually identical.

6.2.3 Feature Engagement

The feature engagement analysis compared average feature usage per account between active and churned customers. Churned customers recorded approximately 522 average usage per account, while active customers recorded approximately 495.

This finding does not support the simple assumption that customers are leaving because they barely use the product. In fact, churned customers showed slightly greater average engagement.

However, this should not be interpreted as higher feature usage causing churn. Instead, it shows that overall feature usage alone does not clearly distinguish active customers from churned customers. As decision-makers, we would investigate which specific features customers use and whether they achieve useful outcomes rather than simply attempting to increase total usage.

6.2.4 Support Resolution Time

Average support resolution time was approximately 35.92 hours for active customers and 35.66 hours for churned customers.

The difference is very small. Therefore, the data does not provide strong evidence that churned customers generally experienced slower ticket resolution.

This finding changes the possible management response. As RavenStack's management team, we would not immediately set a costly organisation-wide objective to reduce average resolution time purely as a churn solution. Other support characteristics may be more important.

6.2.5 System Error Impact

The comparison of churned versus active customers using System Error Analysis shows very little variance. On average, churned customers had a total of about 0.541 errors while active customers had an average of almost 0.571 errors.

Therefore, the existing data does not demonstrate that churned customers experience significantly more system errors than active customers. It does not mean that system performance (i.e. how well things work) is less important; it simply means there appears to be limited ability for System Error Count to differentiate churners from non-churners based on this particular data set.

6.2.6 Support Escalation and Churn

Support escalation produced a more meaningful difference than general resolution time. Accounts associated with escalated support cases recorded a churn rate of approximately 25.32%, compared with 21.72% for accounts without escalation.

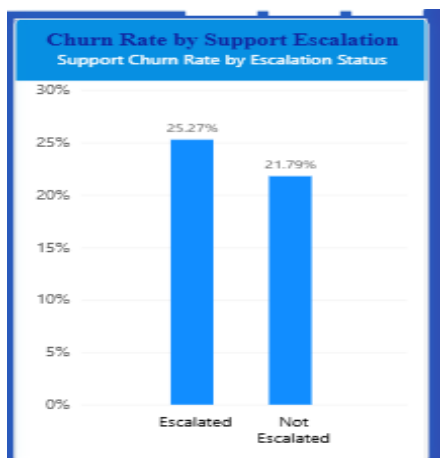


Figure 6. Churn Rate by Support Escalation Status

From a management perspective, this suggests that the seriousness or complexity of customer support problems deserves more attention than average resolution time alone. We would therefore focus on what happens during and after escalated cases, including customer communication, follow-up and confirmation that the underlying problem has been resolved.

Taken together, the diagnostic analysis gives an important management conclusion: not every

suspected churn factor is equally supported by the data. Industry and support escalation show clearer differences, while plan tier, average resolution time, system errors and overall feature engagement do not provide strong evidence of being major churn differentiators on their own.

6.3 Predictive Findings: Customer Risk and Future MRR

6.3.1 Key Churn Influencers

Power BI Key Influencers was used to identify customer characteristics associated with increased churn likelihood. The visual identified two main factors:

- Industry = DevTools: 1.60 times higher likelihood of churn
- Referral Source = Event: 1.51 times higher likelihood of churn



Figure 7. Power BI Key Influencers for Customer Churn

The DevTools result is particularly important because it supports the earlier diagnostic finding. DevTools already had the highest observed industry churn rate at 30.97%, and it was also identified by Power BI as a key churn influencer.

From a management perspective, receiving the same signal through two analytical views increases the importance of investigating this customer segment.

Event referral provides an additional insight that was not obvious from the descriptive acquisition chart alone. Although the descriptive dashboard showed where customers came from, the predictive analysis suggests that Event-acquired customers have a higher likelihood of churn than the comparison group shown by Power BI.

6.3.2 Active Customer Risk Classification

The identified characteristics were used to build a transparent rule-based churn-risk score. DevTools was given 55 points, while Event referral was given 45 points because DevTools was the stronger of the two displayed influencers.

The 390 currently active accounts were classified as follows.

Table 4: Active Accounts by Churn Risk Level

Risk Level	Active Accounts
Low Risk	258
Medium Risk	54
High Risk	78
Total	390

The most significant outcome was the identification of 78 active accounts at "high" risk. These customers have churned (yet) however contain attributes known to be indicative of a greater churn risk.

A high-risk active customer list including account id., account name, industry, referral source, and a churn risk score and risk level can also be seen on the dashboard. This takes the analytical process, and turns it into an operational tool.

We as RavenStack's executive team will utilize this list to determine which customer(s), if any should receive our initial focus; vs. waiting for them to cancel.

Important to note is that the risk score generated by the model is NOT a machine learning based probability. For example, if a customer receives a score of 100, it doesn't indicate there is a 100% chance they will churn. Rather it indicates their account contains both of the risk characteristic(s) that were used in determining the rule based classification.

6.3.3 Three-Month MRR Forecast

The final predictive analysis used RavenStack's historical monthly MRR to create a three-month MRR forecast.

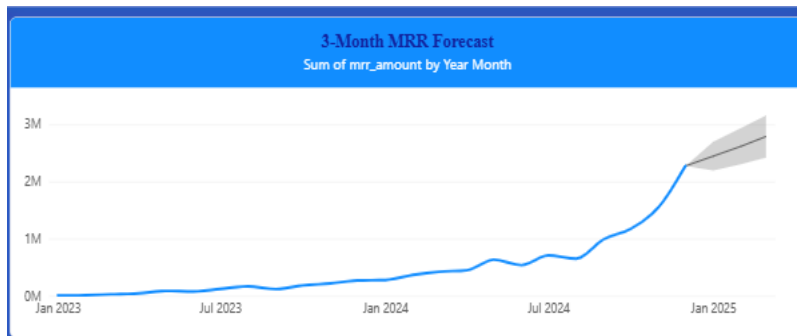


Figure 8. RavenStack Three-Month MRR Forecast

The historical data shows strong MRR growth toward the end of the observed period, and the Power BI forecast projects a continued upward direction over the following three months. The shaded area represents the 95% confidence interval, which shows uncertainty around the projected values.

As RavenStack's management team, we would treat this forecast as a planning tool rather than a guaranteed financial result. The upward direction is encouraging, but future MRR will depend on whether RavenStack can maintain customer acquisition while controlling churn.

7.0 Decision-Making Implications

This study has indicated that RavenStack should adopt a targeted (as opposed to generic) retention policy prior to launching publicly. DevTools is particularly in need of close management consideration due to it having the greatest industrial churn rate and being identified as the most significant influence of churn.

In addition, the support results indicate that management should prioritize escalated issues over time-to-resolution. In regard to the three subscription tiers (Basic, Pro & Enterprise), the lack of difference in the two-churn rates suggests there may be no large scale change required in relation to any of the subscription levels.

Most importantly, the study has identified 78 high risk actively engaged subscribers which will enable RavenStack to transition from a reactive to a pro-active approach in terms of customer retention. Thusly, management can take action before those potential lost customers are actually

gone, they do not need to only respond after the fact.

Finally, although the revenue MRR forecast continues to predict growth, we will continue to monitor performance by measuring both revenue growth and retention moving forward. Therefore, our managerial decisions would be to proceed with growth but strengthen our targeted retention programs.

8.0 Recommendations

Based upon our results, we suggest four specific steps that RavenStack needs to take.

We recommend that RavenStack focus on supporting its most important customers – those who use DevTools. DevTools had the largest churn rate (at 30.97%) and were found to have a 1.60 greater probability of churning based on our Key Influencers analysis. Therefore, it would be helpful for RavenStack to collect input from both active and churned DevTools customers to help determine what issues may exist regarding fit, functionality, or what customers are expecting in terms of their experience with the product.

Secondly, we recommend that RavenStack implement a formalized follow-up procedure when dealing with escalations. Churn rates for escalated cases were significantly larger (25.32%) than for non-escalated cases (21.72%). Thus, after an escalated case has been resolved, the Customer Success team should follow up with the customer to verify that all issues have been completely resolved prior to closing out the case.

Thirdly, we suggest that RavenStack prioritize the 78 high-risk active accounts (as shown by the High Risk Active Customer List) for proactive retention efforts. Since these customers have not churned yet, they provide RavenStack the opportunity to make proactive contact to retain these customers.

Finally, we suggest that management evaluate how to best address potential churn risks of customers acquired via events, since Event Referral was positively correlated to a 1.51 times greater likelihood of churning. We recommend that management assess whether there is some disconnect between the messaging provided to attract customers to attend events versus what those same customers expect once they onboard into the product.

Overall, we recommend that RavenStack develop and implement a focused retention strategy versus a broad-based approach; our research clearly indicates that the risk of churning is not

equally distributed among all customers.

9.0 Limitations

Our analysis has several limitations. First, the dataset contains only 500 accounts from the pilot period, so customer behaviour may change after the wider public launch.

Second, the diagnostic results identify relationships but do not prove causation. For example, support escalation is associated with higher churn, but this does not prove that escalation directly causes customers to leave.

Third, our Churn Risk Score is a rule-based classification informed by Power BI Key Influencers, rather than a separately trained machine-learning probability model. Therefore, the score is used to prioritise customers and should not be interpreted as an exact probability of churn.

Finally, the three-month MRR forecast is based on historical patterns. Actual future revenue may change because of customer behaviour, marketing activities, pricing changes and other business conditions.

10.0 Conclusion

This project demonstrated how business analytics can support RavenStack's customer-retention and revenue decisions. Descriptive analysis identified a 22% overall churn rate, while diagnostic analysis showed that DevTools customers and accounts associated with support escalation require greater attention.

Predictive analysis further identified DevTools and Event referral as churn influencers, classified 78 active accounts as high risk, and provided a three-month MRR forecast. Based on these findings, we recommend targeted retention actions rather than applying the same strategy to all customers.

Overall, the project demonstrates how RavenStack can use data to move from understanding business performance to making focused and evidence-based management decisions.

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Appendix A: Individual Contribution Statement

Table A1. Individual Group Member Contributions

Group Member	Contribution
Hassan Ahmed Abdikadir	Coordinated group activities and task distribution, monitored project progress, and contributed to the development of all three analytical stages: descriptive, diagnostic, and predictive analytics . Also supported dashboard review and interpretation of the findings.
Mohamed Mohamed Hussein	Contributed to dataset preparation, data cleaning, data modelling and relationships in Power BI. Supported the development of dashboard visuals and DAX measures.
Hussein Musab Abdi	Contributed to the development of the descriptive, diagnostic, and predictive analysis , supported the creation of dashboard visuals and measures, and assisted in interpreting the analytical findings.
Mohamed Luqman Hassan	Prepared and organised the final project report and presentation slides , selected and organised dashboard screenshots and visual evidence, formatted the report, and supported preparation for the group presentation.
All Members	Participated in discussing findings, developing recommendations, reviewing the final dashboard and report, and preparing for the project presentation.

Appendix B: Project Dataset and Power BI Files

The complete supporting files used for the RavenStack analytics project are available through the shared folder below. The folder contains the five original Excel datasets used in the analysis and the Power BI (.pbix) project file containing the data model, DAX calculations, and Descriptive, Diagnostic, and Predictive Analytics dashboards.

Project Files Folder:

[\[Click here to access the RavenStack Dataset and Power BI Files\]](#)